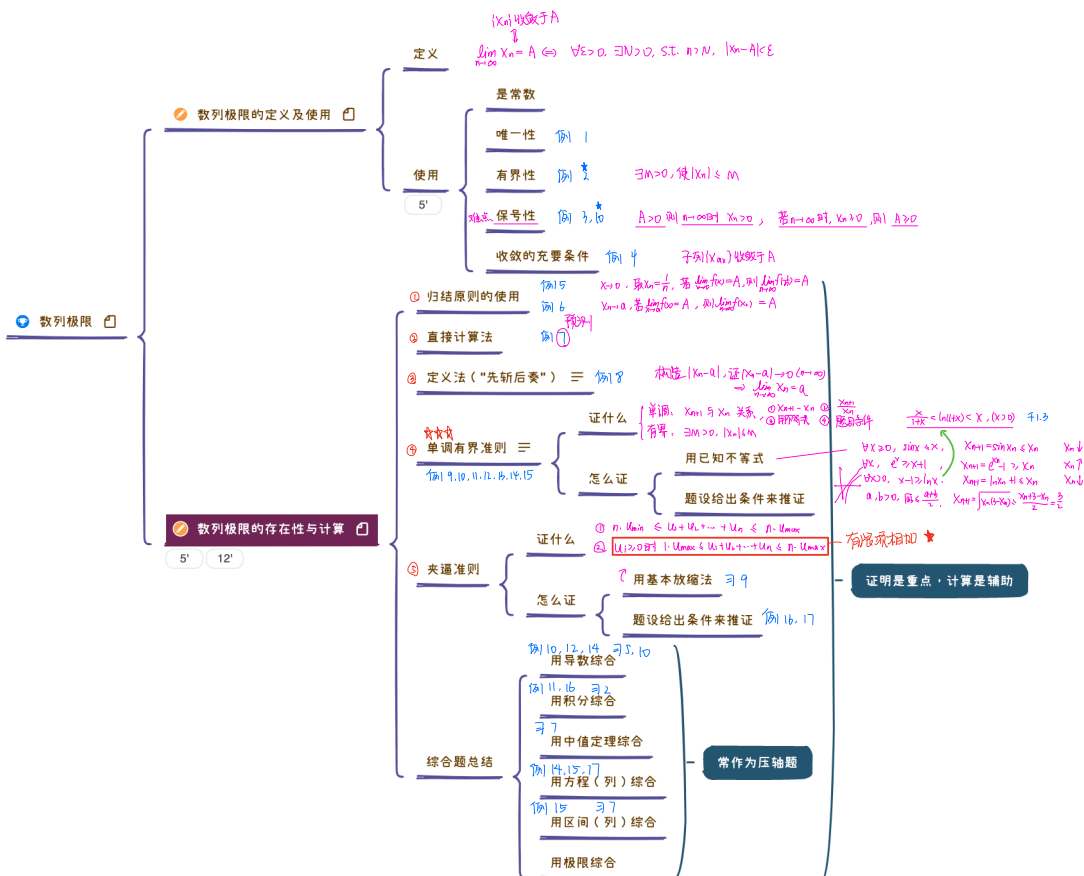


高等数学

数列极限



例2.1 唯一性

例2.1 设数列 $\{x_n\}$ 收敛, 则 (D).

- (A) 当 $\lim_{n \rightarrow \infty} \sin x_n = 0$ 时, $\lim_{n \rightarrow \infty} x_n = 0$
 (B) 当 $\lim_{n \rightarrow \infty} (x_n + \sqrt{|x_n|}) = 0$ 时, $\lim_{n \rightarrow \infty} x_n = 0$
 (C) 当 $\lim_{n \rightarrow \infty} (x_n + x_n^2) = 0$ 时, $\lim_{n \rightarrow \infty} x_n = 0$
 (D) 当 $\lim_{n \rightarrow \infty} (x_n + \sin x_n) = 0$ 时, $\lim_{n \rightarrow \infty} x_n = 0$

$$\text{收敛} \Rightarrow \lim_{n \rightarrow \infty} x_n = A$$

$$A) \sin A = 0 \not\Rightarrow A = 0$$

$$b) A + \sqrt{|A|} = 0 \not\Rightarrow A = 0$$

$$c) A + A^2 = 0 \not\Rightarrow A = 0$$

例2.2 有界性

例2.2 设数列 $\{a_n\}$ 单调减少, $\{b_n\}$ 单调增加, 且 $\lim_{n \rightarrow \infty} (a_n - b_n) = 0$, 则 ().

- (A) $\{a_n\}$ 收敛, $\{b_n\}$ 不收敛
 (B) $\{a_n\}$ 不收敛, $\{b_n\}$ 收敛
 (C) $\{a_n\}, \{b_n\}$ 均收敛, 但 $\lim_{n \rightarrow \infty} a_n \neq \lim_{n \rightarrow \infty} b_n$
 (D) $\{a_n\}, \{b_n\}$ 均收敛, 且 $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n$

突破口

$$\lim_{n \rightarrow \infty} a_n - b_n = 0 \Rightarrow \{a_n - b_n\} \text{ 有界}$$

$$\Rightarrow A \leq a_n - b_n \leq B$$

$$\textcircled{1} a_n \geq A + b_n \geq A + b_1, \{a_n\} \text{ 单减有下界} \Rightarrow \lim_{n \rightarrow \infty} a_n \exists$$

$$\textcircled{2} b_n \leq a_n - A \leq a_1 - A, \{b_n\} \text{ 单增有上界} \Rightarrow \lim_{n \rightarrow \infty} b_n \exists$$

$$0 = \lim_{n \rightarrow \infty} (a_n - b_n) = \lim_{n \rightarrow \infty} a_n - \lim_{n \rightarrow \infty} b_n \Rightarrow \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n \Rightarrow (D)$$

例2.3 保号性

例2.3 已知 $a_n = \sqrt[n]{n} - \frac{(-1)^n}{n} (n = 1, 2, \dots)$, 则 $\{a_n\}$ ().

- (A) 有最大值, 有最小值
 (B) 有最大值, 没有最小值
 (C) 没有最大值, 有最小值
 (D) 没有最大值, 没有最小值

最值是比较出来的

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} e^{\frac{\ln n}{n}} = 1$$

$$a_2 = \sqrt{2} - \frac{1}{2} < 0$$

$$a_1 = 1 + 1 > 0$$

$$\left. \begin{aligned} \lim_{n \rightarrow \infty} (a_n - a_2) > 0 &\Rightarrow \exists N_2 > 0, n > N_2 \text{ 时}, a_n > a_2 \\ \lim_{n \rightarrow \infty} (a_n - a_1) < 0 &\Rightarrow \exists N_1 > 0, n > N_1 \text{ 时}, a_n < a_1 \end{aligned} \right\} N > \max\{N_1, N_2\} \text{ 时} \quad a_2 < a_N < a_1 \Rightarrow (A)$$

例2.4 收敛的必要条件

例2.4 设 $\{x_n\}$ 是数列, 下列命题中不正确的是 (D).

- (A) 若 $\lim_{n \rightarrow \infty} x_n = a$, 则 $\lim_{n \rightarrow \infty} x_{2n} = \lim_{n \rightarrow \infty} x_{2n+1} = a$
 (B) 若 $\lim_{n \rightarrow \infty} x_{2n} = \lim_{n \rightarrow \infty} x_{2n+1} = a$, 则 $\lim_{n \rightarrow \infty} x_n = a$
 (C) 若 $\lim_{n \rightarrow \infty} x_n = a$, 则 $\lim_{n \rightarrow \infty} x_{3n} = \lim_{n \rightarrow \infty} x_{3n+1} = a$
 (D) 若 $\lim_{n \rightarrow \infty} x_{3n} = \lim_{n \rightarrow \infty} x_{3n+1} = a$, 则 $\lim_{n \rightarrow \infty} x_n = a$

例 2.10 保号性

例 2.10 设函数 $f(x) = \ln x + \frac{1}{x}$.

(1) 求 $f(x)$ 的最小值;

(2) 设数列 $\{x_n\}$ 满足 $\ln x_n + \frac{1}{x_{n+1}} < 1$, 证明 $\lim_{n \rightarrow \infty} x_n$ 存在, 并求此极限.

$$1) f'(x) = \frac{1}{x} - \frac{1}{x^2} = \frac{x-1}{x^2} = 0 \Rightarrow x=1 \quad f(1)=1$$

$$2) \text{ 设 } \lim_{n \rightarrow \infty} x_n = A$$

$$\text{则 } \lim_{n \rightarrow \infty} y_n = \lim_{n \rightarrow \infty} \ln x_n + \frac{1}{x_{n+1}} - 1 < 0 \Rightarrow B = \ln A + \frac{1}{A} - 1 < 0 \Rightarrow \ln A + \frac{1}{A} \leq 1$$

例 2.5 归结原则

例 2.5 求 $\lim_{n \rightarrow \infty} n^3 \left(\sin \frac{1}{n} - \frac{1}{2} \sin \frac{2}{n} \right)$.

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\sin \frac{1}{n} - \frac{1}{2} \sin \frac{2}{n}}{\left(\frac{1}{n}\right)^3} &= \lim_{x \rightarrow 0} \frac{\sin x - \frac{1}{2} \sin 2x}{x^3} = \lim_{x \rightarrow 0} \frac{\sin x - \frac{1}{2} \sin 2x}{x^3} = \lim_{x \rightarrow 0} \frac{\frac{1}{6}x^3 - \frac{1}{6}x^3}{x^3} = \frac{1}{6} \end{aligned}$$

例 2.6 归结原则 铺垫

例 2.6 设 $a_1 > 0$, $\{a_n\}$ 满足 $a_{n+1} = \ln(1+a_n)$, $n=1, 2, \dots$.

(1) 证明 $\lim_{n \rightarrow \infty} a_n$ 存在, 并求其值;

(2) 求 $\lim_{n \rightarrow \infty} \frac{a_{n+1} a_n}{a_n - a_{n+1}}$.

【注】(1) 若能大体画出 $y=x$ 与 $y=\ln(1+x)$ 的图形, 如图 2-1 所示, 就会发现方程 $x=\ln(1+x)$ 的根是 $x=0$, 也可以发现, 当 $a_1 > 0$ 时, $\{a_n\}$ 单调减少并收敛于 0. 这种直观的发现虽不能代替证明, 但能够得以发现其中的规律, 为证明提供思路和解.

(2) (*) 处来自归结原则. (**) 处当 $x \rightarrow 0^+$ 时分子 $\ln(1+x) \sim x$, 分母 $x - \ln(1+x) \sim \frac{1}{2}x^2$, 这来自泰勒公式.

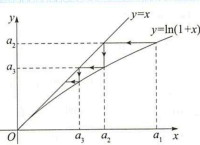


图 2-1

$$\begin{aligned} 1) \quad a_1 > 0. \text{ 设 } a_{k+1} > 0, a_k = \ln(1+a_{k-1}) > 0 &\Rightarrow \{a_n\} \text{ 有下界 } 0 \\ \text{当 } a_n > 0 \text{ 时 } a_{n+1} = \ln(1+a_n) < a_n &\Rightarrow \{a_n\} \text{ 单调减} \\ \Rightarrow a = \ln(1+a) &\Rightarrow a=0 \end{aligned} \quad \left. \begin{array}{l} \lim_{n \rightarrow \infty} a_n = a \end{array} \right\}$$

$$2) \lim_{n \rightarrow \infty} \frac{a_{n+1} a_n}{a_n - a_{n+1}} = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{a_{n+1}} - \frac{1}{a_n}} = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{\ln(1+a_{n+1})} - \frac{1}{a_n}} = \lim_{x \rightarrow 0^+} \frac{1}{\frac{1}{\ln(1+x)} - \frac{1}{x}} = \lim_{x \rightarrow 0^+} \frac{x^2}{x - \ln(1+x)} = 2$$

例 2.7 直接计算法

例 2.7 设 $a_1 = 3, a_{n+1} = a_n^2 + a_n (n=1, 2, \dots)$, 求极限

$$\lim_{n \rightarrow \infty} \left(\frac{1}{1+a_1} + \frac{1}{1+a_2} + \dots + \frac{1}{1+a_n} \right).$$

$$a_{n+1} = a_n(a_n + 1) > a_n \Rightarrow \{a_n\} \uparrow$$

$$\left. \begin{array}{l} \{a_n\} \uparrow \text{ 且有上界 } \Rightarrow \lim_{n \rightarrow \infty} a_n = A \text{ (存在)} \\ \{a_n\} \uparrow \text{ 且无上界 } \Rightarrow \lim_{n \rightarrow \infty} a_n = +\infty \end{array} \right\}$$

$$\textcircled{1} \{a_n\} \text{ 有上界 } \Rightarrow A = A^2 + A \Rightarrow A^2 = 0 \Rightarrow A = 0 \quad \left. \begin{array}{l} \text{矛盾} \end{array} \right\}$$



★ $a_n > 3, \lim_{n \rightarrow \infty} a_n > 3 > 0$

★ $\Rightarrow \textcircled{2} |a_n| \text{ 无上界}, \lim_{n \rightarrow \infty} a_n = +\infty$

$$\frac{1}{a_{n+1}} - \frac{1}{a_n(a_n+1)} = \frac{1}{a_n} - \frac{1}{a_n+1} \Rightarrow \frac{1}{a_{n+1}} = \frac{1}{a_n} - \frac{1}{a_n+1}$$

$$\Rightarrow \text{原} = \frac{1}{a_1} - \frac{1}{a_2} + \frac{1}{a_2} - \frac{1}{a_3} + \dots = \frac{1}{a_1} - \left(\frac{1}{a_n}\right)^0$$

$$\text{故 } \lim_{n \rightarrow \infty} \left(\frac{1}{1+a_1} + \frac{1}{1+a_2} + \dots + \frac{1}{1+a_n}\right) = \frac{1}{a_1} = \frac{1}{3}$$



★ 例 2.8 定义法

例 2.8 设 $f(x)$ 满足

① $a \leq f(x) \leq b, x \in [a, b]$;

② 对 $\forall x, y \in [a, b], |f(x) - f(y)| \leq \frac{1}{2} |x - y|$.

又 $\{x_n\}$ 满足 $a \leq x_1 \leq b, x_{n+1} = \frac{1}{2} [x_n + f(x_n)]$.

(1) 证明 $f(x) = x$ 在 $[a, b]$ 上有唯一解, 记为 c ;

(2) 证明 $\lim_{n \rightarrow \infty} x_n = c$.

先证连续性

$$\forall x, x_0 \in [a, b], |f(x) - f(x_0)| \leq \frac{1}{2} |x - x_0|$$

$$\text{故 } 0 \leq \lim_{x \rightarrow x_0} |f(x) - f(x_0)| \leq \lim_{x \rightarrow x_0} \frac{1}{2} |x - x_0| = 0$$

$$\Rightarrow \lim_{x \rightarrow x_0} |f(x) - f(x_0)| = 0 \Rightarrow \lim_{x \rightarrow x_0} f(x) = f(x_0)$$

(1) 令 $F(x) = f(x) - x$

$$F(a) = f(a) - a \geq 0$$

$$F(b) = f(b) - b \leq 0$$

当 $F(a) = 0$ 时 c 取 a 即可

当 $F(b) = 0$ 时 c 取 b 即可

由零点定理 $\exists c \in (a, b)$ 使 $F(c) = 0$

反证法 设 $d \in [a, b], d \neq c$, 使 $F(d) = 0$

$$0 = F(d) - F(c) = f(d) - d - f(c) + c$$

$$\Rightarrow |f(d) - f(c)| = |d - c|$$

$$\text{又 } |f(d) - f(c)| \leq \frac{1}{2} |d - c| \quad \text{矛盾}$$

故 c 唯一

$$c = f(c)$$

$$c + c = f(c) + c$$

$$\uparrow c = \frac{1}{2} (c + f(c))$$

$$(2) \text{ 柯西 } |x_{n+1} - c| = \left| \frac{1}{2} (x_n + f(x_n)) - \frac{1}{2} (c + f(c)) \right|$$

$$= \left| \frac{1}{2} (x_n - c) + \frac{1}{2} (f(x_n) - f(c)) \right|$$

$$|a+b| \leq |a| + |b| \quad \left\{ \begin{aligned} &\leq \frac{1}{2} |x_n - c| + \frac{1}{2} |f(x_n) - f(c)| \\ &\leq \frac{1}{2} |x_n - c| + \frac{1}{2} |x_n - c| = \frac{3}{4} |x_n - c| \\ &\leq \left(\frac{3}{4}\right)^n |x_1 - c| \end{aligned} \right.$$

$$n \rightarrow \infty \text{ 时 } 0 < |x_{n+1} - c| \leq \lim_{n \rightarrow \infty} \left(\frac{3}{4}\right)^n |x_1 - c| = 0$$

$$\therefore \lim_{n \rightarrow \infty} x_{n+1} = c$$



★★★ 例 2.9 (新)

例 2.9 已知 $(2 + \sqrt{2})^n = A_n + \sqrt{2} B_n, A_n, B_n$ 为整数, $n = 1, 2, 3, \dots$, 求 $\lim_{n \rightarrow \infty} \frac{A_n}{B_n}$.

$$(2 + \sqrt{2})^n = A_n + \sqrt{2} B_n$$

$$\star \text{ 递推 } A_{n+1} + \sqrt{2} B_{n+1} = (2 + \sqrt{2})^{n+1} = (2 + \sqrt{2})^n (2 + \sqrt{2})$$

$$= (A_n + \sqrt{2} B_n) (2 + \sqrt{2}) \leftarrow \begin{aligned} &2A_n + 2\sqrt{2}B_n + \sqrt{2}A_n + 2B_n \\ &= (2A_n + 2B_n) + \sqrt{2}(A_n + 2B_n) \end{aligned}$$

$$\Rightarrow \begin{cases} A_{n+1} = 2A_n + 2B_n \\ B_{n+1} = A_n + 2B_n \end{cases}$$

$$\frac{A_{n+1}}{B_{n+1}} = \frac{2\frac{A_n}{B_n} + 2}{\frac{A_n}{B_n} + 2}$$

$$\text{令 } X_n = \frac{A_n}{B_n}, X_{n+1} = \frac{2X_n + 2}{X_n + 2} = 2 - \frac{2}{X_n + 2}$$

$$x_{n+1} - x_n = \frac{2}{x_{n+1} + 2} - \frac{2}{x_n + 2} = \frac{2(x_n - x_{n-1})}{(x_{n+1} + 2)(x_n + 2)}$$

$\Rightarrow x_{n+1} - x_n$ 与 $x_n - x_{n-1}$ 同号 $\Rightarrow x_n$ 单调.

由单调有界准则 $\Rightarrow \lim_{n \rightarrow \infty} x_n$ 存在, 记为 A

$$A = 2 - \frac{1}{A+2} \Rightarrow A = \sqrt{2} \quad \text{即} \quad \lim_{n \rightarrow \infty} \frac{A_n}{B_n} = \sqrt{2}$$

由题设 $x_n > 0 \Rightarrow x_{n+1} < 2$



★★★

例 2.11

例 2.11 设 $x_1 = 1, x_n = \int_0^1 \min\{x, x_{n-1}\} dx, n = 2, 3, \dots$, 证明 $\lim_{n \rightarrow \infty} x_n$ 存在并求其极限值.

paper tiger

$$x_2 = \int_0^1 \min\{x, 1\} dx = \frac{1}{2}$$

$$x_3 = \int_0^1 \min\{x, \frac{1}{2}\} dx = \int_0^{\frac{1}{2}} x dx + \int_{\frac{1}{2}}^1 \frac{1}{2} dx = \frac{3}{8}$$

$$\Rightarrow x_n = \int_0^{x_{n-1}} x dx + \int_{x_{n-1}}^1 x_{n-1} dx = \frac{x_{n-1}^2}{2} + x_{n-1}(1 - x_{n-1}) = x_{n-1} - \frac{x_{n-1}^2}{2} > 0$$

$$\text{且 } x_n = x_{n-1} - \frac{x_{n-1}^2}{2} < x_{n-1}$$

$$\therefore \{x_n\} \text{ 单调有下界} \Rightarrow A = A - \frac{A^2}{2} \Rightarrow A = 0 \quad \text{即} \quad \lim_{n \rightarrow \infty} x_n = 0$$



★★★

例 2.12

例 2.12 (1) 证明方程 $x = 2\ln(1+x)$ 在 $(0, +\infty)$ 内有唯一实根 ξ ;

(2) 对于 (1) 中的 ξ , 任取 $x_1 > \xi$, 定义 $x_{n+1} = 2\ln(1+x_n), n = 1, 2, \dots$, 证明 $\lim_{n \rightarrow \infty} x_n = \xi$.

$$(1) f(x) = -2\ln(1+x) + x$$

$$f'(x) = \frac{-2}{1+x} + 1 \stackrel{?}{=} 0 \Rightarrow x = 1 \text{ 唯一驻点}$$

当 $0 < x < 1$ 时 $f'(x) < 0$, $x > 1$ 时 $f'(x) > 0$

$$\because f(0) = 0 \quad f(1) = 1 - 2\ln 2 < 0 \quad \lim_{x \rightarrow +\infty} f(x) = +\infty$$

$\Rightarrow (0, 1)$ 无实根 $(1, +\infty)$ 有 $f(\eta) = 0$ (零点定理)

$$\Rightarrow \underbrace{\xi = 2\ln(1+\xi)}_{\text{第一问的宝贵财富}}, 1 < \xi < +\infty.$$

$$(2) x_1 > \xi, f(x_1) = x_1 - 2\ln(1+x_1) > 0 \Rightarrow x_1 > 2\ln(1+x_1)$$

$$\star x_2 = 2\ln(1+x_1) < x_1$$

$$\because g(x) = \ln(1+x) \uparrow \quad \left. \begin{array}{l} x_1 > \xi \\ x_1 > \xi \end{array} \right\} \Rightarrow \underbrace{2\ln(1+x_1)}_{x_2} > \underbrace{2\ln(1+\xi)}_{\xi} \Rightarrow x_2 > \xi$$

$$\left. \begin{array}{l} x_1 > x_2 > \xi \end{array} \right\}$$

$$x_{n+1} = 2\ln(1+x_n)$$

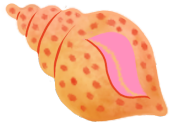
$$\text{设 } x_{n-1} > x_n > \xi$$

$$\left. \begin{array}{l} x_n > 2\ln(1+x_n) = x_{n+1} \\ 2\ln(1+\xi) = \xi \end{array} \right\} \Rightarrow x_n > x_{n+1} > \xi$$

故 $\{x_n\}$ 单调有下界

$$\Rightarrow \lim_{n \rightarrow \infty} x_n \text{ 存在, 记为 } A$$

$$\Rightarrow A = 2\ln(1+A) \Rightarrow A = \xi$$



例 2.13

设函数 $f(x)$ 连续, $\forall a_1, a_{n+1} = f(a_n), n = 1, 2, 3, \dots$. 关于下列两个结论:

- ① 若 $f(x)$ 严格单增且有上界, 则数列 $\{a_n\}$ 收敛;
- ② 若 $f(x)$ 严格单减且有界, 则数列 $\{a_n\}$ 收敛.

正确的选项是().

- (A) 仅 ① 正确
(C) ①② 都正确

- (B) 仅 ② 正确
(D) ①② 都错误

$$\lim_{x \rightarrow +\infty} f(x) = A \Rightarrow \lim_{n \rightarrow \infty} f(a_n) = A$$

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} f(a_n)$$

不要人为加戏, 说了上界就不要人为加下界

反例: 取 $f(x) = \begin{cases} \arctan x, & x > 0 \\ 2x, & x < 0 \end{cases}$

$$a_1 = -1, a_2 = -2, \dots$$

反例: 取 $f(x) = -\frac{1}{x} \arctan x$

$$a_1 = 1 \Rightarrow a_2 = -1 \Rightarrow a_3 = 1 \dots$$



例 2.14 方程列

设 $f_n(x) = \cos x + \cos^2 x + \dots + \cos^n x (n = 1, 2, \dots)$. 证明:

- (1) 对于每个 n , 方程 $f_n(x) = 1$ 在 $[0, \frac{\pi}{3})$ 内有且仅有一个实根 x_n ;

(2) $\lim_{n \rightarrow \infty} x_n = \frac{\pi}{3}$.

	函数列	方程列	数列
(1)	$f_1(x) = \cos x$	$= 1$	x_1
	$f_2(x) = \cos x + \cos^2 x$	$= 1$	x_2
	$\vdots$	$\vdots$	$\vdots$

$$\text{令 } F_n(x) = f_n(x) - 1, n = 1, 2, \dots$$

$$F_n(0) = f_n(0) - 1 = n - 1 > 0$$

$$F_n(\frac{\pi}{3}) = f_n(\frac{\pi}{3}) - 1 = \frac{1 - (\frac{1}{2})^n}{1 - \frac{1}{2}} - 1 = -(\frac{1}{2})^n < 0$$

零点定理 至少一个 x_n

$$\therefore F'_n(x) = f'_n(x) = -\sin x + 2\cos x(-\sin x) + \dots + n\cos^n x(-\sin x)$$

$$= -\sin x(1 + 2\cos x + \dots + n\cos^{n-1} x) < 0, x \in (0, \frac{\pi}{3})$$

$\therefore$ 仅有一个 x_n

$$(2) \quad f_n(x_n) = 1 \Rightarrow \cos x_n + \cos^2 x_n + \dots + \cos^n x_n = 1$$

$$f_{n+1}(x_{n+1}) = 1 \Rightarrow \cos x_{n+1} + \cos^2 x_{n+1} + \dots + \cos^{n+1} x_{n+1} = 1$$

$$\text{令 } g(x) = \cos x + \cos^2 x + \dots + \cos^n x \downarrow$$

$$\therefore g(x_n) > g(x_{n+1}) \Rightarrow x_n < x_{n+1} \Rightarrow \{x_n\} \uparrow$$

$$\text{由 (1)} \quad x_n < \frac{\pi}{3} \Rightarrow \lim_{n \rightarrow \infty} x_n \text{ 存在} = a$$

$$f_n(x_n) = 1 \quad \text{则} \quad \cos x_n + \dots + \cos^n x_n = 1 \Rightarrow \frac{\cos x_n (1 - \cos^{n+1} x_n)}{1 - \cos x_n} = 1$$

($\lim_{n \rightarrow \infty} (1 - \cos^n x_n) \neq 0 \Rightarrow \lim_{n \rightarrow \infty} (1 - \frac{1}{n})^n = e^{-1}$)

$$\Rightarrow 0 < x_n < \frac{\pi}{3} \Rightarrow 0 < \cos x_n < \cos x_1 < 1$$

$$\downarrow \quad \downarrow \quad \downarrow$$

要开一个数隔开

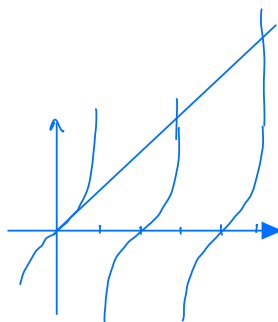
$$n \rightarrow \infty \quad \frac{\cos a}{1 - \cos a} = 1 \Rightarrow a = \frac{\pi}{3}$$



例2.15 区间列

例 2.15 (1) 证明方程 $\tan x = x$ 在 $(n\pi, n\pi + \frac{\pi}{2})$ 内存在实根 $\xi_n, n = 1, 2, 3, \dots$;

(2) 求极限 $\lim_{n \rightarrow \infty} (\xi_{n+1} - \xi_n)$.



11) $\hat{=} f(x) = \tan x - x, x \in (\pi, \pi + \frac{3}{2})$
 $f(\pi) = -\pi < 0$
 $\lim_{x \rightarrow (\pi+\frac{3}{2})} \tan x - x = +\infty > 0$ $\Rightarrow \exists \eta_n \in (\pi, \pi + \frac{\pi}{2})$ 使 $\tan \eta_n = \eta_n$

$$1^{\circ} \quad n \rightarrow \infty \Rightarrow g_n \rightarrow +\infty$$

$$1^\circ \quad n \rightarrow \infty \Rightarrow f_n \rightarrow +\infty$$

$$2^\circ \quad \frac{\pi}{2} < \varphi_{n+1} - \varphi_n < \frac{3}{2}\pi$$

$$\Rightarrow \lim_{n \rightarrow \infty} \tan(g_{n+1} - g_n) = 0 \quad , \quad g_{n+1} - p_n \in \left(\frac{\pi}{2}, \frac{3}{2}\pi\right)$$

$$\Rightarrow g_{n+1} - g_n = \pi$$



★ 2.9

2.9 求 $f(x) = \lim_{n \rightarrow \infty} \sqrt[n]{1 + x^n + \left(\frac{x^2}{2}\right)^n} \ (x \geq 0)$ 的表达式.

用极限定义函数

$$1 + X^n + \left(\frac{X^2}{2}\right)^n$$

$$\left. \begin{aligned} \textcircled{1} \quad x \in (0, 1) \text{ 时} \quad & \sqrt[n]{1 \cdot 1^n} \leq \sqrt[n]{1^n + x^n + \left(\frac{x}{2}\right)^n} \leq \sqrt[n]{3 \cdot 1^n} \\ & \quad \quad \quad \parallel \quad \quad \quad \quad \quad \quad \quad \quad \quad \parallel \\ & \quad \quad \quad 1 \quad \quad \quad \quad \quad \quad \quad \quad \quad 1 \\ \textcircled{2} \quad x \in (1, 2) \text{ 时} \quad & \sqrt[n]{1 \cdot x^n} \leq \sqrt[n]{1^n + x^n + \left(\frac{x}{2}\right)^n} \leq \sqrt[n]{3 \cdot x^n} \\ & \quad \quad \quad \parallel \quad \quad \quad \quad \quad \quad \quad \quad \quad \parallel \\ & \quad \quad \quad x \quad \quad \quad \quad \quad \quad \quad \quad \quad x \\ \textcircled{3} \quad x \in [2, +\infty) \text{ 时} \quad & \sqrt[n]{1 \cdot \left(\frac{x}{2}\right)^n} \leq \sqrt[n]{1^n + x^n + \left(\frac{x}{2}\right)^n} \leq \sqrt[n]{3 \cdot \left(\frac{x}{2}\right)^n} \\ & \quad \quad \quad \parallel \quad \quad \quad \quad \quad \quad \quad \quad \quad \parallel \\ & \quad \quad \quad \frac{x}{2} \quad \quad \quad \quad \quad \quad \quad \quad \quad \frac{x}{2} \end{aligned} \right\} f(x) = \begin{cases} 1, & x \in (0, 1) \\ x, & x \in (1, 2) \\ \frac{x}{2}, & x \in [2, +\infty) \end{cases}$$



*** 例 2.16

(2023)

例 2.16 $\lim_{n \rightarrow \infty} \int_0^1 (n+1)x^n \ln(1+x) dx = (\quad).$

(A)ln 2

(B)1

 $(C)e^2$

(D) $+\infty$

$$\int_0^1 \ln(1+x) \, dx = x^{n+1} \cdot \ln(1+x) \Big|_0^1 - \int_0^1 \frac{x^{n+1}}{x+1} \, dx = \ln 2 - \int_0^1 \frac{x^{n+1}}{x+1} \, dx$$

$$0 \leq \int_0^1 \frac{x^{n+1}}{x+1} dx \leq \int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{1}{n+1} \quad (A)$$



☆☆☆

例 2.17

例 2.17 设 $a_n = \int_0^1 x^n \sqrt{1-x^2} dx (n=0,1,2,\dots)$.

(1) 证明: 数列 $\{a_n\}$ 单调减少, 且 $a_n = \frac{n-1}{n+2} a_{n-2} (n=2,3,\dots)$;

(2) 求 $\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}}$ $\downarrow$ < 1

$$\begin{aligned} (1) \quad a_{n+1} - a_n &= \int_0^1 (x^{n+1} - x^n) \sqrt{1-x^2} dx \\ &= \int_0^1 x^n (x-1) \sqrt{1-x^2} dx < 0 \end{aligned}$$

$\Rightarrow \{a_n\} \downarrow$.

令 $x = \sin t$

$$a_n = \int_0^{\frac{\pi}{2}} \sin^n t \cdot \cos^2 t dt = \int_0^{\frac{\pi}{2}} \sin^n t dt - \int_0^{\frac{\pi}{2}} \sin^{n+2} t dt = b_n - b_{n+2} = b_n - \frac{n+1}{n+2} b_n = \frac{1}{n+2} b_n$$

$$a_{n-2} = \int_0^{\frac{\pi}{2}} \sin^{n-2} t dt - \int_0^{\frac{\pi}{2}} \sin^n t dt = b_{n-1} - b_n = \frac{n}{n-1} b_n - b_n = \frac{1}{n-1} b_n$$

$$\Rightarrow a_n = \frac{n-1}{n+2} a_{n-2}$$

$$(2) \quad \frac{a_n}{a_{n-1}} = \frac{n-1}{n+2} \frac{a_{n-2}}{a_{n-1}}$$

$\{a_n\} \downarrow$

$$\Rightarrow \frac{n-1}{n+2} < \frac{a_n}{a_{n-1}} = \frac{n-1}{n+2} \frac{a_{n-2}}{a_{n-1}} < 1$$

$\downarrow$
 $n \rightarrow \infty$ $\rightarrow$ 1 $\leftarrow$ 1

习 2.1

2.1 已知数列 $\{a_n\}$ 单调, 下列结论正确的是 (B, D).

X (A) $\lim_{n \rightarrow \infty} (e^{a_n} - 1)$ 存在

(B) $\lim_{n \rightarrow \infty} \frac{1}{1+a_n^2}$ 存在

X (C) $\lim_{n \rightarrow \infty} \sin a_n$ 存在

(D) $\lim_{n \rightarrow \infty} \frac{1}{1-a_n^2}$ 存在

	$+\infty$	$-\infty$	A	-A	0
XA	X	✓	✓	✓	✓
XC	X	X	✓	✓	✓
B	✓	✓	✓	✓	✓
D	✓	✓	✓	✓	✓

$$a_n = 1 - \frac{1}{n+1} \quad \lim_{n \rightarrow \infty} \frac{1}{1-a_n^2} = \lim_{n \rightarrow \infty} \frac{1}{1 - \left(\frac{n}{n+1}\right)^2} = \infty$$

习 2.2

2.2 设 $a_n = \frac{3}{2} \int_0^{\frac{1}{n+1}} x^{n+1} \sqrt{1+x^n} dx$, 则极限 $\lim_{n \rightarrow \infty} a_n =$ (B).

(A) $(1+e)^{\frac{2}{3}} + 1$

(B) $(1+e^{-1})^{\frac{2}{3}} - 1$

(C) $(1+e^{-1})^{\frac{2}{3}} + 1$

(D) $(1+e)^{\frac{2}{3}} - 1$

$$\begin{aligned} \frac{3}{2} \int_0^{\frac{1}{n+1}} n x^{n+1} \sqrt{1+x^n} dx &= \frac{3}{2} \int_0^{\frac{1}{n+1}} (1+x^n)^{\frac{1}{2}} d(1+x^n) = \int_0^{\frac{1}{n+1}} d(1+x^n)^{\frac{3}{2}} = \left[1 + \left(\frac{n}{n+1}\right)^{\frac{3}{2}}\right] - 1 \\ \lim_{n \rightarrow \infty} \left[1 + \left(1 - \frac{1}{n+1}\right)^{\frac{3}{2}}\right] - 1 &= (1+e^{-1})^{\frac{2}{3}} - 1 \end{aligned}$$

习2.3

2.3 设 $a_1 = 1, a_2 = 2, a_{n+2} = \frac{2a_n a_{n+1}}{a_n + a_{n+1}} (n = 1, 2, \dots)$.

(1) 求 $b_n = \frac{1}{a_{n+1}} - \frac{1}{a_n}$ 的表达式;

(2) 求 $\sum_{k=1}^n b_k$ 和 $\lim_{n \rightarrow \infty} a_n$.

$$\begin{aligned} (1) \quad a_{n+2} &= \frac{2}{\frac{1}{a_n} + \frac{1}{a_{n+1}}} \Rightarrow \frac{1}{a_n} + \frac{1}{a_{n+1}} = \frac{2}{a_{n+2}} \\ \text{设 } C_n &= \frac{1}{a_n} \quad \text{则 } C_n + C_{n+1} = 2C_{n+2}, \quad C_1 = 1, \quad C_2 = \frac{1}{2}, \quad C_3 = \frac{1}{3}, \quad C_4 = \frac{1}{4}, \dots \\ b_n &= C_{n+1} - C_n = \left(-\frac{1}{2}\right)^{n-1} \\ (2) \quad \sum_{i=1}^n b_i &= \frac{(-\frac{1}{2})(1 - (-\frac{1}{2})^n)}{1 - (-\frac{1}{2})} = \frac{(-\frac{1}{2})^n - 1}{3} \\ C_{n+1} &= C_n + b_n = C_{n-1} + b_{n-1} + b_n = C_1 + \sum_{i=1}^n b_i = \frac{(-\frac{1}{2})^n + 2}{3} \\ \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \frac{3}{2 + (-\frac{1}{2})^n} = \frac{3}{2} \end{aligned}$$

习2.4

2.4 已知 $x_1 = \frac{1}{2}, 2x_{n+1} + x_n^2 = 1, \lim_{n \rightarrow \infty} x_n$.

证 $\lim_{n \rightarrow \infty} x_n$ 存在, 则 $2a + a^2 = 1 \Rightarrow a = \sqrt{2} - 1$

$$0 \leq |x_{n+1} - (\sqrt{2} - 1)| < \left| \frac{1 - x_n^2}{2} - \frac{1 - a^2}{2} \right| = \frac{1}{2} |x_n^2 - a^2| = \frac{1}{2} |x_n + a| \cdot |x_n - a|$$

$$\text{当 } n > 1 \text{ 时, } x_n < \frac{1}{2} \Rightarrow \frac{x_n + a}{2} < \frac{1}{2} \Rightarrow 0 < |x_{n+1} - (\sqrt{2} - 1)| < \frac{1}{2} |x_n - a| < \left(\frac{1}{2}\right)^n |x_1 - a|$$

习2.5

2.5 (1) 设 $f(x) = x + \ln(2-x)$, 求 $f(x)$ 的最大值;

(2) 设 $x_1 = \ln 2, x_n = \sum_{i=1}^{n-1} \ln(2-x_i), n = 2, 3, \dots$, 证明 $\lim_{n \rightarrow \infty} x_n$ 存在并求其极限值.

$$(1) \quad f'(x) = 1 - \frac{1}{2-x} \stackrel{?}{=} 0, \quad x = 1$$

$$f(x)_{\max} = f(1) = 1$$

$$x + \ln(2-x) \leq 1 \quad \ln(2-x) \leq 1-x$$

$$\begin{aligned} (2) \quad x_{n+1} &= \sum_{i=1}^n \ln(2-x_i) = x_n + \ln(2-x_n) \leq 1, \text{ 有上界} \\ \because x \leq 1 \text{ 时 } \ln(2-x) \geq 1, &\Rightarrow f(x) \geq x, \quad x_{n+1} = f(x_n) \geq x_n. \quad \{x_n\} \uparrow \text{ 极限存在} \\ A &= A + \ln(2-A) \Rightarrow A = 1 \end{aligned}$$

习2.6

2.6 设 $x_{n+1} = \cos x_n, n = 1, 2, \dots, x_1 = \cos x$, 证明 $\lim_{n \rightarrow \infty} x_n$ 存在且其极限是方程 $\cos x - x = 0$ 的根.

$$\text{令 } f(x) = \cos x - x$$

$$f(0) = 1 > 0, \quad f\left(\frac{\pi}{2}\right) = \frac{1}{2} - \frac{\pi}{2} < 0$$

$$\therefore f(x) \text{ 的根在 } (0, \frac{\pi}{2}) \text{ 内} \Rightarrow a \in (0, \frac{\pi}{2}), \cos a = a$$

由于 $\cos x$ 是偶函数, $-1 \leq x_1 \leq 1$, 则 $0 \leq x_n \leq 1, n > 1$, 又 $1 < \frac{\pi}{2}$, 故 $0 \leq x_n < \frac{\pi}{2}$

$$\therefore |x_{n+1} - a| = |\cos x_n - \cos a| = 2 \left| \sin \frac{x_n + a}{2} \sin \frac{x_n - a}{2} \right|$$

$$\leq 2 \sin \frac{\pi}{2} \left| \sin \frac{x_n - a}{2} \right| \leq 2 \sin \frac{\pi}{2} \cdot \left| \frac{x_n - a}{2} \right| = \frac{\pi}{2} |x_n - a|$$

$$\Rightarrow |x_{n+1} - a| \leq \dots \leq \left(\frac{\pi}{2}\right)^n |x_1 - a|$$

$$n \rightarrow \infty \text{ 时 } \left(\frac{\pi}{2}\right)^n |x_1 - a| \rightarrow 0, \Rightarrow \lim_{n \rightarrow \infty} x_{n+1} = a$$

$$\lim_{n \rightarrow \infty} x_n = a$$

$$\text{帅+帅=帅哥, } (\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2})$$

$$\text{帅-帅=哥帅, } (\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2})$$

$$\text{哥+哥=哥哥, } (\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2})$$

$$\text{哥-哥=负嫂嫂, } (\cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2})$$

知乎 @Aries

积化和差只需反过来说, 即:

• 帅哥=帅+帅

• 哥帅=帅-帅

• 哥哥=哥+哥

• 负嫂嫂=哥-哥

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

$$\cos \alpha \sin \beta = \frac{1}{2} [\sin(\alpha + \beta) - \sin(\alpha - \beta)]$$

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha + \beta) + \cos(\alpha - \beta)]$$

$$\sin \alpha \sin \beta = -\frac{1}{2} [\cos(\alpha + \beta) - \cos(\alpha - \beta)]$$

知乎 @Aries

习 2.7

2.7 (1) 证明对任意的正整数 n , 都有 $\frac{1}{n+1} < \ln\left(1 + \frac{1}{n}\right) < \frac{1}{n}$ 成立;

(2) 设 $a_n = 1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln n$ ($n = 1, 2, \dots$), 证明数列 $\{a_n\}$ 收敛.

$$(1) \ln\left(1 + \frac{1}{n}\right) = \ln \frac{n+1}{n} = \ln(n+1) - \ln n$$

$$\ln(n+1) - \ln n = \ln\left(\frac{n+1}{n}\right) = \frac{1}{\xi}, \xi \in (n, n+1)$$

$$\therefore \frac{1}{n+1} < \ln\left(1 + \frac{1}{n}\right) = \frac{1}{\xi} < \frac{1}{n}$$

$$(2) a_n = 1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln n$$

$$a_{n+1} = 1 + \frac{1}{2} + \dots + \frac{1}{n+1} - \ln(n+1)$$

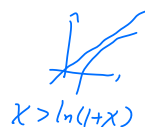
$$a_{n+1} - a_n = \frac{1}{n+1} - \ln \frac{n+1}{n} < 0 \Rightarrow \{a_n\} \downarrow$$

$$a_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} - \ln n$$

$$> \ln(1+1) + \ln(1+\frac{1}{2}) + \dots + \ln(1+\frac{1}{n}) - \ln n$$

$$= \ln(2) + \ln 3 - \ln 2 + \ln 4 - \ln 3 + \dots + \ln(n+1) - \ln n - \ln n$$

$$= \ln(n+1) - \ln n > 0 \quad \text{有下界}$$



习 2.8

2.8 设数列 $\{x_n\}$ 满足: $x_1 > 0, x_n e^{x_{n+1}} = e^{x_n} - 1$ ($n = 1, 2, \dots$), 证明 $\{x_n\}$ 收敛, 并求 $\lim_{n \rightarrow \infty} x_n$.

$$\textcircled{1} x_1 > 0, \quad x_n \cdot e^{x_{n+1}} = e^{x_n} - 1 > x_n$$

$$\Rightarrow e^{x_{n+1}} > 1 \Rightarrow x_{n+1} > 0 \quad \text{有下界}$$

$$\textcircled{2} x_1 > 0, \quad e^{x_1} = \frac{e^{x_2} - 1}{x_1} = \frac{e^{x_2} - e^0}{x_1} = \frac{(x_1 - 0)e^{\xi}}{x_1} = e^{\xi}, \xi \in (0, x_1)$$

$$\therefore e^{x_1} > e^{\xi} \quad \therefore e^{x_2} = e^{\xi} < e^{x_1}$$

$$\text{假设 } 0 < x_{n+1} < x_n \text{ 则 } e^{x_{n+1}} = e^{\eta} < e^{x_n}, \eta \in (0, x_n)$$

$$\Rightarrow \{x_n\} \downarrow \text{ 有下界 } \Rightarrow x_n \text{ 收敛}$$

$$\text{设 } \lim_{n \rightarrow \infty} x_n = a$$

$$a e^a = e^a - 1 \Rightarrow a = 0$$

习 2.10

2.10 设 $f_0(x)$ 是 $[0, +\infty)$ 上连续的严格单调增加函数, 函数 $f_1(x) = \int_0^x f_0(t) dt$,

(1) 补充定义 $f_1(x)$ 在 $x=0$ 处的值, 使得补充定义后的函数 (仍记为 $f_1(x)$) 在 $[0, +\infty)$ 上连续;

(2) 在 (1) 的条件下, 证明 $f_1(x) < f_0(x)$ ($x > 0$), 且 $f_1(x)$ 也是 $[0, +\infty)$ 上连续的严格单调增加函数;

$$(1) \lim_{x \rightarrow 0} f_1(x) = \lim_{x \rightarrow 0} \frac{\int_0^x f_0(t) dt}{x} = \lim_{x \rightarrow 0} \frac{f_0(x)}{1} = f_0(0)$$

$$\text{补充定义 } f_1(0) = f_0(0)$$

$$(2) f_1(x_0) = \frac{\int_0^{x_0} f_0(t) dt}{x_0} < \frac{\int_0^{x_0} f_0(x_0) dt}{x_0} = f_0(x_0)$$

$$\text{设 } a < b \text{ 则 } f_1(a) = \frac{\int_0^a f_0(t) dt}{a} < \frac{\int_0^a f_0(b) dt}{b} < \frac{\int_0^b f_0(b) dt}{b} = f_1(b)$$



$$f_1(x) = \frac{\int_0^x f_0(t) dt}{x} = \frac{x \cdot f_0(\eta)}{x} = f_0(\eta), \eta \in (0, x)$$

$$\because f_0(x) \uparrow \Rightarrow f_1(x) = f_0(\eta) < f_0(x)$$

$$f_1'(x) = \frac{x f_0'(x) - \int_0^x f_0(t) dt}{x^2} = \frac{x f_0'(x) - x f_0(\eta)}{x^2} > 0 \Rightarrow f_1(x) \uparrow$$

(3) 令 $f_n(x) = \frac{\int_0^x f_{n-1}(t) dt}{x}, n = 1, 2, 3, \dots$, 证明: 对任意的 $x > 0$, 极限 $\lim_{n \rightarrow \infty} f_n(x)$ 存在.

$$\begin{aligned} f_n(x) &= \frac{\int_0^x f_{n-1}(t) dt}{x} \\ \text{设 } a < b \quad f_n(a) &= \frac{\int_0^a f_{n-1}(t) dt}{a} < \frac{\int_0^a f_{n-1}(t) dt + \int_a^b f_{n-1}(t) dt}{b} < \frac{\int_0^b f_{n-1}(t) dt}{b} = f_n(b) \\ \Rightarrow f_n(x) \uparrow \quad \text{设 } f_{n-1}(x) \uparrow \\ \text{① } f_n(x) &= \frac{\int_0^x f_{n-1}(t) dt}{x} = f_{n-1}(\eta) \\ \Rightarrow f_n(x) &= f_{n-1}(\eta) < f_{n-1}(x) \\ \Rightarrow \{f_n(x)\} \downarrow \\ \text{② } f_n(x) &> \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{\int_0^x f_{n-1}(t) dt}{x} = \lim_{n \rightarrow \infty} f_{n-1}(x) = \dots = f_0(0) \\ \Rightarrow \text{单调有下界} \Rightarrow \lim_{n \rightarrow \infty} f_n(x) \text{ 存在} \end{aligned}$$

2.11

2.11 设 $f_n(x) = 1 - (1 - \cos x)^n$.

(1) 证明方程 $f_n(x) = \frac{1}{2}$ 在 $(0, \frac{\pi}{2})$ 内仅有一根 $x_n, n = 1, 2, 3, \dots$;

(2) 求 $\lim_{n \rightarrow \infty} f_n(\arccos \frac{1}{n})$;

(3) 设 $x_n \in (0, \frac{\pi}{2})$ 满足 $f_n(x_n) = \frac{1}{2}$, 证明 $\lim_{n \rightarrow \infty} x_n = \frac{\pi}{2}$.

$$\begin{aligned} (1) \text{ 设 } F_n(x) &= f_n(x) - \frac{1}{2} \\ F_n(0) &= f_n(0) - \frac{1}{2} = \frac{1}{2} \\ F_n(\frac{\pi}{2}) &= f_n(\frac{\pi}{2}) - \frac{1}{2} = -\frac{1}{2} \\ F_n'(x) &= -n(1 - \cos x)^{n-1} < 0 \Rightarrow \text{单调, } \Rightarrow \text{唯一} \end{aligned}$$

$$\begin{aligned} (2) \lim_{n \rightarrow \infty} f_n(\arccos \frac{1}{n}) \\ = \lim_{n \rightarrow \infty} 1 - (1 - \frac{1}{n})^n = \lim_{x \rightarrow 0} 1 - (1 - x)^{\frac{1}{x}} = \lim_{x \rightarrow 0} 1 - e^{\frac{1}{x}(-x)} = 1 - e^{-1} \end{aligned}$$

$$\begin{aligned} (3) \lim_{n \rightarrow \infty} f_n(\arccos \frac{1}{n}) &= 1 - e^{-1} > \frac{1}{2} \\ \text{根据极限保号性 } \exists N, n > N \text{ 时 } f_n(\arccos \frac{1}{n}) &> \frac{1}{2} = f_n(x_n) \\ \text{由(1) } f_n(x) \text{ 在 } (0, \frac{\pi}{2}) \text{ 内严格单调减少, 故 } \arccos \frac{1}{n} &< x_n < \frac{\pi}{2} \\ \lim_{n \rightarrow \infty} \arccos \frac{1}{n} &= \frac{\pi}{2} \Rightarrow \lim_{n \rightarrow \infty} x_n \end{aligned}$$

2.1.1

1. 已知数列 $\{x_n\}$ 满足 $x_{n+2} - \frac{4}{3}x_{n+1} + \frac{1}{3}x_n = 0, n = 1, 2, 3, \dots$, 且 $x_1 = 1, x_2 = 2$, 则 $\{x_n\}$ 收敛于 (C).

(A) 1

(B) -1

(C) $\frac{5}{2}$

(D) $-\frac{5}{2}$

$$x_{n+2} = \frac{4}{3}x_{n+1} - \frac{1}{3}x_n = \frac{1}{3}x_{n+1} - \frac{1}{3}x_n + x_{n+1} = (\frac{1}{3}x_{n+1} - \frac{1}{3}x_n) + (\frac{1}{3}x_n - \frac{1}{3}x_{n-1}) + \dots + (\frac{1}{3}x_2 - \frac{1}{3}x_1) + (\frac{1}{3}x_1 - \frac{1}{3}x_0) + x_2$$

$$A = \frac{1}{3}A + \frac{5}{3} \Rightarrow \frac{5}{3} = \frac{2}{3}A \Rightarrow A = \frac{5}{2} = \frac{1}{3}x_{n+1} + 2 - \frac{1}{3} = \frac{1}{3}x_{n+1} + \frac{5}{3}$$

习2.1.2

2. 设 $x_n = 1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+n}$, 则 $\lim_{n \rightarrow \infty} x_n =$ _____.

$$\frac{1}{1+2+3+\dots+n} = \frac{2}{(n+1)n} = 2\left(\frac{1}{n} - \frac{1}{n+1}\right)$$

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n} - \frac{1}{n+1}\right) = \lim_{n \rightarrow \infty} \frac{2n}{n+1} = 2$$



习2.1.3

3. 设 $x_n = \left(1 + \frac{1}{n}\right)\left(1 + \frac{2}{n}\right) \dots \left(1 + \frac{n}{n}\right)$, 则 $\lim_{n \rightarrow \infty} x_n =$ _____.

$$x_n = \prod_{i=1}^n \frac{n+i}{n} = \frac{n^2+n}{n^2}$$

$$e^2 = \lim_{n \rightarrow \infty} \left(\frac{n^2+n}{n^2}\right)^n < \lim_{n \rightarrow \infty} x_n < \lim_{n \rightarrow \infty} \left(\frac{n^2+n}{n^2}\right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

$$x_n = \prod_{i=1}^n \frac{n+i}{n^2}$$

$\frac{x}{x+1} < (n+1)x < x$

$$\sum_{i=1}^n \frac{i}{n^2} < \ln x_n = \sum_{i=1}^n \ln \frac{n+i}{n^2} = \sum_{i=1}^n \ln \left(1 + \frac{i}{n}\right) < \sum_{i=1}^n \frac{i}{n^2}$$

左右 $n \rightarrow \infty$ 时, 左右 $\rightarrow \frac{n(n+1)}{2n^2} = \frac{1}{2}$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \ln \frac{n+i}{n^2} = \frac{1}{2}$$

$$\lim_{n \rightarrow \infty} \prod_{i=1}^n \left(1 + \frac{i}{n}\right) = e^{\frac{1}{2}}$$

习2.1.4

4. 设 $x > 0, n$ 为正整数, 记 $f(x) = \lim_{n \rightarrow \infty} [x^{\frac{1}{n}} - x^{\frac{1}{n+1}}]$, 则 $f'(x) =$ _____.

$$f(x) = \lim_{n \rightarrow \infty} n^2 \left[x^{\frac{1}{n}} - x^{\frac{1}{n+1}} \right]$$

$$= \lim_{n \rightarrow \infty} n^2 \left[e^{\frac{1}{n} \ln x} - e^{\frac{1}{n+1} \ln x} \right] \quad \text{设 } f(x) = e^{\frac{1}{x} \ln x}$$

$$= \lim_{n \rightarrow \infty} n^2 \left[(1-1) \cdot \frac{\ln x}{x} \cdot x^{\frac{1}{n}} \cdot \left(-\frac{1}{n^2}\right) \right], \quad f \in (n, n+1)$$

$$= \lim_{n \rightarrow \infty} \frac{\ln x}{x} e^{\frac{\ln x}{n}} = \frac{\ln x}{x}$$

$$f'(x) = \frac{1 - \ln x}{x^2}$$

习2.1.5

5. 求 $\lim_{n \rightarrow \infty} n^2 \left(\arctan \frac{a}{n} - \arctan \frac{a}{n+1} \right), a > 0$.

$$\lim_{n \rightarrow \infty} n^2 \left([n - (n+1)] \cdot \frac{1}{1 + \left(\frac{a}{n}\right)^2} \cdot \frac{-a}{n^2} \right) = \lim_{n \rightarrow \infty} \frac{a}{1 + \left(\frac{a}{n}\right)^2} = a$$

习2.1.6

6. 设 $F(x, y) = \frac{f(y-x)}{2x}, F(1, y) = \frac{y^2}{2} - y + 5, x_0 > 0, x_1 = F(x_0, 2x_0), \dots, x_{n+1} = F(x_n, 2x_n),$
 $n = 1, 2, \dots$, 证明 $\lim_{n \rightarrow \infty} x_n$ 存在, 并求该极限.

$$F(1, y) = \frac{f(y-1)}{2} = \frac{y^2}{2} - y + 5 \Rightarrow f(y-1) = y^2 - 2y + 10 = (y-1)^2 + 9$$

$$\Rightarrow f(x) = x^2 + 9 \Rightarrow F(x, 2x) = \frac{x^2 + 9}{2x}$$

$$\Rightarrow x_{n+1} = \frac{x_n^2 + 9}{2x_n}$$

$$x_{n+1} - x_n = \frac{x_n^2 - 4x_n + 9}{2x_n} = \frac{9 - 3x_n^2}{2x_n} = \frac{9}{2} \frac{1}{x_n} - \frac{3}{2} x_n$$

$$f(x) = \frac{9-3x^2}{2x} \Rightarrow f'(x) = \frac{-12x^2 - (9-3x^2)2}{4x^2} = \frac{-18-6x^2}{4x^2} < 0. \text{ 令 } f(x)=0 \Rightarrow x=\sqrt{3}$$

$$f(x) = \frac{x^2+9}{4x} \quad f'(x) = \frac{8x^2 - (x^2+9)4}{16x^2} = \frac{4x^2-36}{16x^2} = \frac{1}{4} - \frac{9}{4x^2} \stackrel{?}{=} 0 \Rightarrow x=3$$

$$x_1 = \frac{x_0^2+9}{2x_0}, \dots, x_{n+1} = \frac{x_n^2+9}{2x_n} = \frac{1}{2}(x_n + \frac{9}{x_n}) \Rightarrow \sqrt{x_n \cdot \frac{9}{x_n}} = 3 \Rightarrow \{x_n\} \text{ 有下界}$$

$$\frac{x_{n+1}}{x_n} = \frac{1}{2}(1 + \frac{9}{x_n^2}) \leq \frac{1}{2}(1 + \frac{9}{3^2}) = 1 \Rightarrow \{x_n\} \downarrow \Rightarrow \text{极限存在.}$$

$$\text{设 } A = \frac{A^2+9}{2A} \Rightarrow A=3$$

42.1.7

7. 设 $x_1 > 0, x_{n+1} = \ln(x_n + 1) (n=1, 2, \dots)$, 求:

$$(1) \lim_{n \rightarrow \infty} \left(\frac{1}{x_n} - \frac{1}{x_{n+1}} \right);$$

$$(2) \lim_{n \rightarrow \infty} \left(\frac{x_n}{x_{n+1}} \right)^{\frac{1}{x_n}}.$$

$$1 + x + x^2 + \dots = \frac{1}{1-x}$$

$$1 - x + x^2 - \dots = \frac{1}{1+x}$$

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \dots = \ln(1+x)$$

$$x - \frac{x^2}{2} + x^2$$

$$(1) \quad x_{n+1} = \ln(x_n + 1) < x_n, \quad x_n \downarrow \quad \left. \begin{array}{l} x_1 > 0 \Rightarrow x_2 = \ln(x_1 + 1) > 0 \Rightarrow x_n > 0 \Rightarrow \text{有上界} \end{array} \right\} \text{ 极限存在}$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{x_n} - \frac{1}{x_{n+1}} \right) = \lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{x_n x_{n+1}} = \lim_{n \rightarrow \infty} \frac{\ln(x_n + 1) - x_n}{x_n \ln(x_n + 1)} = \lim_{x \rightarrow 0^+} \frac{\ln(x+1) - x}{x \ln(x+1)} = \lim_{x \rightarrow 0^+} \frac{-\frac{x}{2}}{x^2} = -\frac{1}{2}$$

$$(2) \lim_{n \rightarrow \infty} \left(\frac{x_n}{\ln(x_n + 1)} \right)^{\frac{1}{x_n}} = \lim_{x \rightarrow 0^+} e^{\frac{\ln \frac{x}{\ln(x+1)}}{x}} = \lim_{x \rightarrow 0^+} e^{\frac{\ln(1+x)}{x} \cdot \frac{\ln(x) - \frac{x}{1+x}}{\ln^2(1+x)}} = \lim_{x \rightarrow 0^+} e^{\frac{\ln(1+x) - \frac{x}{1+x}}{x^2}} = \lim_{x \rightarrow 0^+} e^{\frac{(1+x)\ln(1+x) - x}{x^2}} = e^{\frac{1}{2}}$$

42.1.8

8. 证明下列结论:

(1) 设比值极限 $\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}} = q$, 若 $q < 1$, 则 $\lim_{n \rightarrow \infty} a_n = 0$, 若 $q > 1$, 则 $\lim_{n \rightarrow \infty} a_n = \infty$;

(2) 设根值极限 $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = q$, 若 $q < 1$, 则 $\lim_{n \rightarrow \infty} a_n = 0$, 若 $q > 1$, 则 $\lim_{n \rightarrow \infty} a_n = \infty$.

$$(1) \quad \frac{|a_n|}{|a_{n-1}|} = q < 1 \rightarrow \{|a_n|\} \downarrow \text{ 且 } |a_n| \geq 0 \Rightarrow \lim_{n \rightarrow \infty} |a_n| = 0 \Rightarrow \lim_{n \rightarrow \infty} a_n = 0$$

$$\frac{|a_n|}{|a_{n-1}|} = q > 1 \rightarrow \{|a_n|\} \uparrow \text{ 若 } |a_n| < A, \text{ 则 } \lim_{n \rightarrow \infty} |a_n| \text{ 存在, 则 } \lim_{n \rightarrow \infty} \frac{|a_n|}{|a_{n-1}|} = 1, \text{ 矛盾}$$

$$\therefore |a_n| < A \text{ 不存在 } \therefore |a_n| \text{ 无界 } \Rightarrow \lim_{n \rightarrow \infty} a_n = \infty$$

$$(2) \quad \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = q < 1 \text{ 时 取 } r, \text{ 使得 } q < r < 1, \text{ 由极限保号性: } \exists N, n > N \text{ 时}$$

$$\sqrt[n]{|a_n|} < r, |a_n| < r^n \Rightarrow 0 < \lim_{n \rightarrow \infty} a_n < \lim_{n \rightarrow \infty} r^n = 0 \Rightarrow \lim_{n \rightarrow \infty} a_n = 0$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = q > 1 \text{ 取 } r, \text{ 使得 } 1 < r < q, \text{ 由极限保号性: } \exists N, n > N \text{ 时}$$

$$\sqrt[n]{|a_n|} > r, |a_n| > r^n \Rightarrow \lim_{n \rightarrow \infty} r^n = \infty \Rightarrow \lim_{n \rightarrow \infty} a_n = \infty$$

千2.9

9. 设 x_0 为任意正数, $x_n = \frac{2022}{2023}x_{n-1} + \frac{1}{x_{n-1}^{2022}}$ ($n = 1, 2, \dots$), 证明 $\lim_{n \rightarrow \infty} x_n$ 存在, 并求 $\lim_{n \rightarrow \infty} x_n$.

$$x_n - x_{n-1} = \frac{1}{x_{n-1}^{2022}} - \frac{x_{n-1}}{2023} = \frac{2023 - x_{n-1}^{2023}}{2023 \cdot x_{n-1}^{2023}}$$

$$x_n = \frac{1}{2023} \left(2022 \cdot x_{n-1} + \frac{2023}{x_{n-1}^{2022}} \right) \geq \left(x_{n-1}^{2022} \cdot \frac{2023}{x_{n-1}^{2023}} \right)^{\frac{1}{2023}} = (2023)^{\frac{1}{2023}} \quad (n \geq 1)$$

$\Rightarrow \{x_n\}$ 有下界

$$x \because x_n \geq (2023)^{\frac{1}{2023}}$$

$$\Rightarrow \frac{x_n}{x_{n-1}} = \frac{2022}{2023} + \frac{1}{x_{n-1}^{2023}} \leq \frac{2022}{2023} + \frac{1}{2023} = 1 \quad (n \geq 2) \quad \Rightarrow \lim_{n \rightarrow \infty} x_n \text{ 存在}$$

$$\begin{aligned} \text{令 } \lim_{n \rightarrow \infty} x_n = a &\Rightarrow a = \frac{2022}{2023}a + \frac{1}{a^{2022}} \\ \Rightarrow a &= (2023)^{\frac{1}{2023}}, \quad \lim_{n \rightarrow \infty} x_n = (2023)^{\frac{1}{2023}} \end{aligned}$$